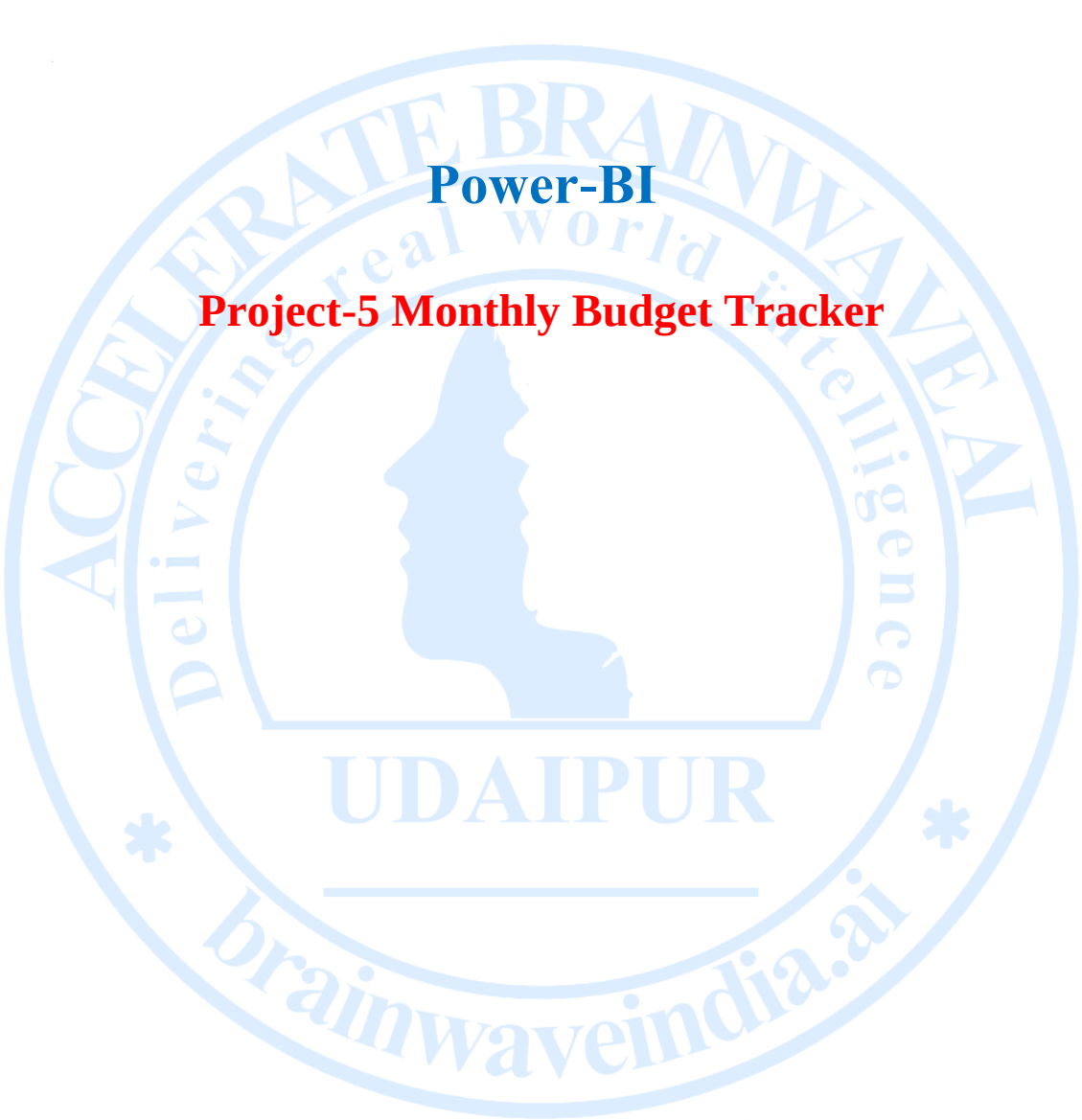




Power-BI

Project-5 Monthly Budget Tracker

Project -5: Monthly Budget Tracker

Goal:

Track budget vs actual spending by category, calculate overspending, and visualize total usage.

Data Fields:

- ❖ Category
- ❖ Budget Amount
- ❖ Actual Expense

STEP 1: Open Excel

- ❖ Launch **Microsoft Excel**
- ❖ Start a **new blank workbook**

STEP 2: Enter Budget Data in Excel

Type the following data starting from **cell A1**:

Category	Budget Amount	Actual Expense
Rent	15000	15000
Groceries	5000	6000
Transport	3000	2500
Utilities	2000	2200
Entertainment	4000	5000
Education	8000	7500
Misc	2000	3000

Double-check column headers:

Category, Budget Amount, Actual Expense

STEP 3: Save the Excel File

Follow these steps:

1. Click **File** in the top-left of Excel.
2. Select **Save As** (or **Browse**).
3. Choose a location (like **Desktop** or **Documents**).
4. In the **File name**, type:
MonthlyBudget.xlsx
5. Ensure **Save as type** is:
Excel Workbook (*.xlsx)
6. Click **Save**

Your budget file is now ready for Power BI.

STEP 4 (Quick Recap): Load Data into Power BI

1. Open **Power BI Desktop**
2. Click **Home > Get Data > Excel**
3. Select MonthlyBudget.xlsx and click **Open**
4. In the Navigator, select the table and click **Load**

Your data is now in Power BI!

STEP 5: Create a Bar Chart for Budget vs Actual

We'll compare **budgeted vs actual expenses per category** using a bar chart.

Follow these steps:

- ❖ Go to **Report View** (white canvas)
- ❖ In the **Visualizations** panel, click **Clustered Column Chart** (vertical bars)
- ❖ A blank chart appears

Add fields to the chart:

- ❖ Drag **Category** to the **X-axis**
- ❖ Drag **Budget Amount** to **Values**
- ❖ Drag **Actual Expense** to **Values**

Now you'll see side-by-side bars comparing budget and actual expenses for each category.

STEP 6: Add KPI Cards & Overspending Calculation

A. Create a Card for Total Budget

1. Click on a blank area of the canvas.
2. In the **Visualizations** panel, click the **Card** visual (□).
3. Drag **Budget Amount** into the **Values** section.
4. It will show the **total budget**.
5. Rename the card title to:
"Total Budget"

B. Create a Card for Total Actual Expense

1. Insert another **Card**.
2. Drag **Actual Expense** into the **Values** section.
3. It will show total actual spending.
4. Rename the title to:
"Total Spent"

C. Create a Measure for Overspending

1. Click on the table name (e.g., `MonthlyBudget`) in the Fields pane.
2. Go to the ribbon and click **New Measure**.
3. Enter this formula:

Overspending = SUM('MonthlyBudget'[Actual Expense]) - SUM('MonthlyBudget'[Budget Amount])

4. Press **Enter**.

D. Create a Card for Overspending

1. Add another **Card** visual.
2. Drag `Overspending` into the **Values** section.
3. Rename the card title to:
"**Overspending**"

If the number is **positive**, it means you're over budget.

If it's **negative**, you're under budget.

STEP 7: Add Pie Chart and/or Table View

A. Create a Pie Chart of Actual Expenses by Category

1. Click on a **blank area** of the canvas.
2. In the **Visualizations** panel, click the **Pie Chart** icon .
3. Drag `Category` to **Legend**.
4. Drag `Actual Expense` to **Values**.
5. Title it: "**Spending by Category**"

You'll now see how your expenses are distributed.

B. (Optional) Create a Table View

1. Click on a **blank space** on the canvas.
2. Choose the **Table** visual from the Visualizations panel (grid icon).

3. Drag in:

- ✓ Category
- ✓ Budget Amount
- ✓ Actual Expense

Now you have a clean view of all data.

STEP 8: Add Slicers (Filters)

A. Add a Slicer for Category

1. Click on a blank space in the canvas.
2. In the **Visualizations** panel, click the **Slicer** icon (filter/list).
3. Drag the `Category` field into the slicer.

Now, selecting a category will filter all visuals (charts, cards, etc.) to show only that category's data.

B. (Optional) Add a Slicer for Budget Amount Range

1. Add another slicer to the canvas.
2. Drag the `Budget Amount` field into it.
3. Power BI will create a **slider** that lets you filter data based on budget value.

You now have a fully interactive dashboard!

STEP 9: Format and Polish the Dashboard

A. Rename All Visual Titles

1. Click on each visual one by one.
2. In the **Visualizations** panel, go to the **Format (paint roller)** tab.
3. Expand the **Title** section:
 - ✓ Make sure **Title** is ON
 - ✓ Enter clear titles like:

- ❖ "Budget vs Actual"
- ❖ "Total Budget"
- ❖ "Total Spent"
- ❖ "Overspending"
- ❖ "Spending by Category"
- ❖ "Filter by Category"

B. Turn on Data Labels (for Charts)

1. Click on your **bar** or **pie** chart.
2. Go to **Format > Data Labels**.
3. Toggle it **ON**.

Now users can see actual values on the chart.

C. Apply a Visual Theme

1. Go to the top menu: **View > Themes**
2. Choose a theme like:

- ❖ "Colorblind Safe"
- ❖ "Minimal"
- ❖ "Blue Yellow"

It instantly applies clean and consistent colors to all visuals.

D. Align & Resize Visuals

1. Drag and drop visuals to arrange them in a grid layout.
2. Resize cards and charts to avoid overlap.
3. Use **Snap to grid** or **Gridlines** from the View tab to align them evenly.

Your dashboard should now be visually polished and easy to read.

STEP 10: Save and Export Your Dashboard

A. Save the Power BI Project File

1. Click **File** (top-left corner).
2. Choose **Save As**.
3. Pick a folder (e.g., Desktop or Documents).
4. Enter a file name:
MonthlyBudgetTracker.pbix
5. Click **Save**

This saves your entire dashboard so you can reopen and edit it anytime in Power BI.

B. Export as PDF (Optional)

1. Click **File > Export > Export to PDF**
2. Power BI generates a PDF of your dashboard
3. Save the file to share it with others

C. (Optional) Publish Online

If you'd like to publish and share online:

1. Click **Home > Publish**
2. Sign in to your Microsoft/Power BI account
3. Select **My Workspace** (or a custom one)
4. Share the link with others via Power BI Service